

Dattansh: Data Analytics

TEAM : ALL FOR ONE

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PROBLEM STATEMENT

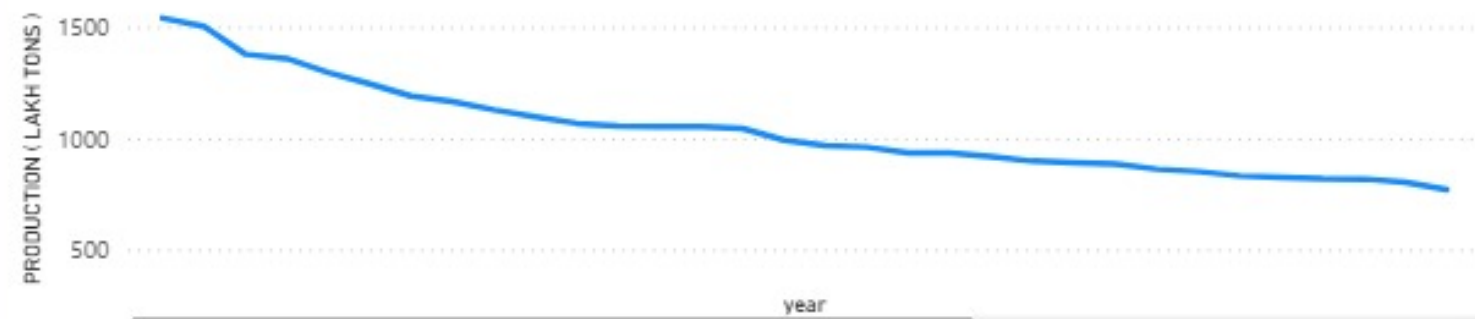
How can historical trajectories, policy shifts, and volatility in India's rice economy—specifically across Minimum Support Price (MSP), cultivation area, yield, and production—be converted into data-driven insights to mitigate supply risks, optimize procurement budgets, and direct capacity planning for the coming season?

[Excel File Link](#)

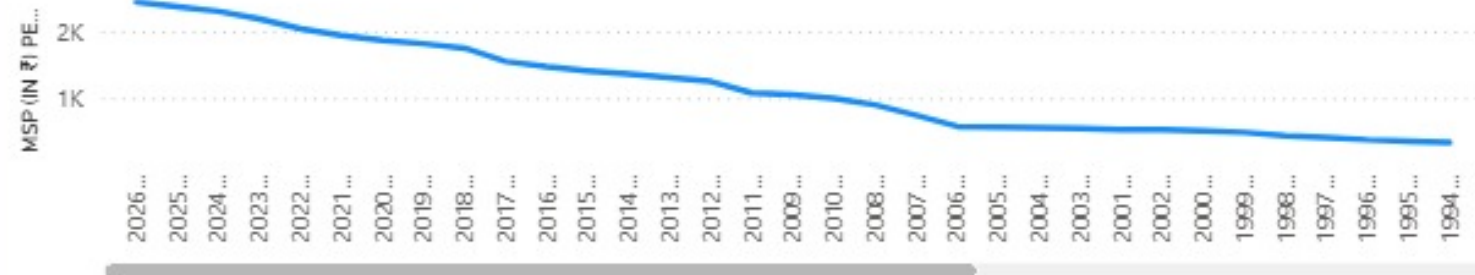
Year	price per 100 kg	Area	Yield	Production	Starting Year	Calculated Production	Production Difference	MSP YoY %	Area YoY %	Yield YoY %	Production YoY %		
2026-27	₹ 2,441.00				2026			3.04%					
2025-26	₹ 2,369.00	528	2915	1540	2025	1539.12	0.88	3.00%	2.92%	-0.48%	2.53%		
2024-25	₹ 2,300.00	513	2929	1502	2024	1502.58	-0.58	5.36%	7.32%	1.63%	9.00%		
2023-24	₹ 2,183.00	478	2882	1378	2023	1377.60	0.40	7.01%	0.00%	1.55%	1.47%		
2022-23	₹ 2,040.00	478	2838	1358	2022	1356.56	1.44	5.15%	3.24%	1.43%	4.86%		
2021-22	₹ 1,940.00	463	2798	1295	2021	1295.47	-0.47	3.85%	1.09%	2.98%	4.10%		
2020-21	₹ 1,868.00	458	2717	1244	2020	1244.39	-0.39	2.92%	4.81%	-0.18%	4.63%		
2019-20	₹ 1,815.00	437	2722	1189	2019	1189.51	-0.51	3.71%	-1.13%	3.18%	2.06%		
2018-19	₹ 1,750.00	442	2638	1165	2018	1166.00	-1.00	12.90%	0.91%	2.41%	3.28%		
2017-18	₹ 1,550.00	438	2576	1128	2017	1128.29	-0.29	5.44%	-0.45%	3.29%	2.83%		
2016-17	₹ 1,470.00	440	2494	1097	2016	1097.36	-0.36	4.26%	1.15%	3.92%	5.08%		
2015-16	₹ 1,410.00	435	2400	1044	2015	1044.00	0.00	3.68%	-0.91%	0.42%	-1.04%		
2014-15	₹ 1,360.00	439	2390	1055	2014	1049.21	5.79	3.82%	-0.23%	-1.40%	-1.12%		
2013-14	₹ 1,310.00	440	2424	1067	2013	1066.56	0.44	4.80%	2.80%	-1.50%	1.43%		
2012-13	₹ 1,250.00	428	2461	1052	2012	1053.31	-1.31	15.74%	-2.73%	2.84%	-0.09%		
2011-12	₹ 1,080.00	440	2393	1053	2011	1052.92	0.08	8.00%	2.56%	6.88%	9.69%		
2010-11	₹ 1,000.00	429	2239	960	2010	960.53	-0.53	-4.76%	2.39%	5.36%	7.74%		
2009-10	₹ 1,050.00	419	2125	891	2009	890.38	0.63	16.67%	-7.91%	-2.43%	-10.15%		
2008-09	₹ 900.00	455	2178	992	2008	990.99	1.01	20.81%	3.64%	-1.09%	2.59%		
2007-08	₹ 745.00	439	2202	967	2007	966.68	0.32	28.45%	0.23%	3.33%	3.53%		
2006-07	₹ 580.00	438	2131	934	2006	933.38	0.62	1.75%	0.23%	1.38%	1.74%		
2005-06	₹ 570.00	437	2102	918	2005	918.57	-0.57	1.79%	4.30%	5.95%	10.47%		
2004-05	₹ 560.00	419	1984	831	2004	831.30	-0.30	1.82%	-1.64%	-4.48%	-6.10%		
2003-04	₹ 550.00	426	2077	885	2003	884.80	0.20	3.77%	3.40%	19.09%	23.26%		
2002-03	₹ 530.00	412	1744	718	2002	718.53	-0.53	0.00%	-8.24%	-16.11%	-23.04%		
2001-02	₹ 530.00	449	2079	933	2001	933.47	-0.47	3.92%	0.45%	9.36%	9.75%		
2000-01	₹ 510.00	447	1901	850	2000	849.75	0.25	4.08%	-1.11%	-4.28%	-5.24%		
1999-00	₹ 490.00	452	1986	897	1999	897.67	-0.67	11.36%	0.89%	3.38%	4.18%		
1998-99	₹ 440.00	448	1921	861	1998	860.61	0.39	6.02%	2.99%	1.11%	4.36%		
1997-98	₹ 415.00	435	1900	825	1997	826.50	-1.50	9.21%	0.23%	0.96%	0.98%		
1996-97	₹ 380.00	434	1882	817	1996	816.79	0.21	5.56%	1.40%	4.73%	6.10%		
1995-96	₹ 360.00	428	1797	770	1995	769.12	0.88	5.88%	0.00%	-5.97%	-5.87%		
1994-95	₹ 340.00	428	1911	818	1994	817.91	0.09	9.68%	0.71%	1.22%	1.87%		
1993-94	₹ 310.00	425	1888	803	1993	802.40	0.60	14.81%	1.67%	8.26%	10.15%		
1992-93	₹ 270.00	418	1744	728	1992	728.99	-0.99	7.74%	-2.11%	-0.40%	-2.41%		
1991-92	₹ 230.00	427	1751	747	1991	747.68	-0.68	12.20%	0.00%	0.63%	0.54%		
1990-91	₹ 205.00	427	1740	743	1990	742.98	0.02	10.81%	1.18%	-0.29%	0.95%		
1989-90	₹ 185.00	422	1745	736	1989	736.39	-0.39	15.63%	1.20%	3.32%	4.40%		
1988-89	₹ 160.00	417	1689	705	1988	704.31	0.69	6.67%	7.47%	15.29%	23.90%		
1987-88	₹ 150.00	388	1465	569	1987	568.42	0.58	2.74%	-5.83%	-0.41%	-6.11%		
1986-87	₹ 146.00	412	1471	606	1986	606.05	-0.05	2.82%	0.24%	-5.22%	-5.02%		
1985-86	₹ 142.00	411	1552	638	1985	637.87	0.13	3.65%	-0.24%	9.53%	9.43%		
1984-85	₹ 137.00	412	1417	583	1984	583.80	-0.80	3.79%	0.00%	-2.75%	-3.00%		
1983-84	₹ 132.00	412	1457	601	1983	600.28	0.72	8.20%	7.50%	18.36%	27.60%		
1982-83	₹ 122.00	383	1231	471	1982	471.47	-0.47	6.09%	-5.90%	-5.89%	-11.63%		
1981-82	₹ 115.00	407	1308	533	1981	532.36	0.64	9.52%	1.24%	-2.10%	-0.56%		
1980-81	₹ 105.00	402	1336	536	1980	537.07	-1.07	10.53%	2.03%	24.39%	26.71%		
1979-80	₹ 95.00	394	1074	423	1979	423.16	-0.16	11.76%	-2.72%	-19.13%	-21.38%		
1978-79	₹ 85.00	405	1328	538	1978	537.84	0.16	10.39%	0.50%	1.53%	2.09%		
1977-78	₹ 77.00	403	1308	527	1977	527.12	-0.12	4.05%	4.68%	20.11%	25.78%		
1976-77	₹ 74.00	385	1089	419	1976	419.27	-0.26	0.00%	-2.53%	-11.82%	-13.96%		
1975-76	₹ 74.00	395	1235	487	1975	487.83	-0.82				0.00%		

India's Rice Economy Has Evolved — But What Is Driving Supply Growth?

PRODUCTION by YEAR



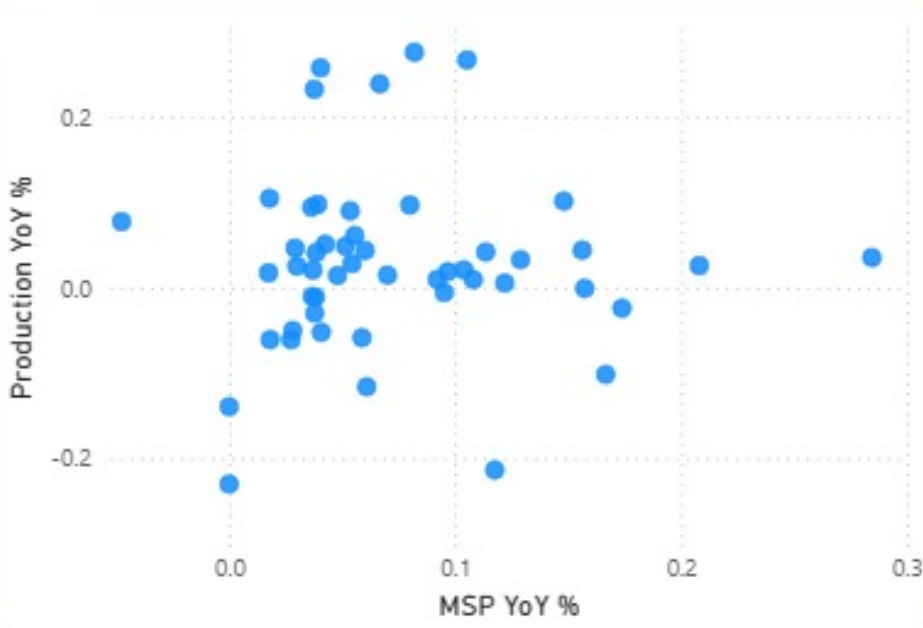
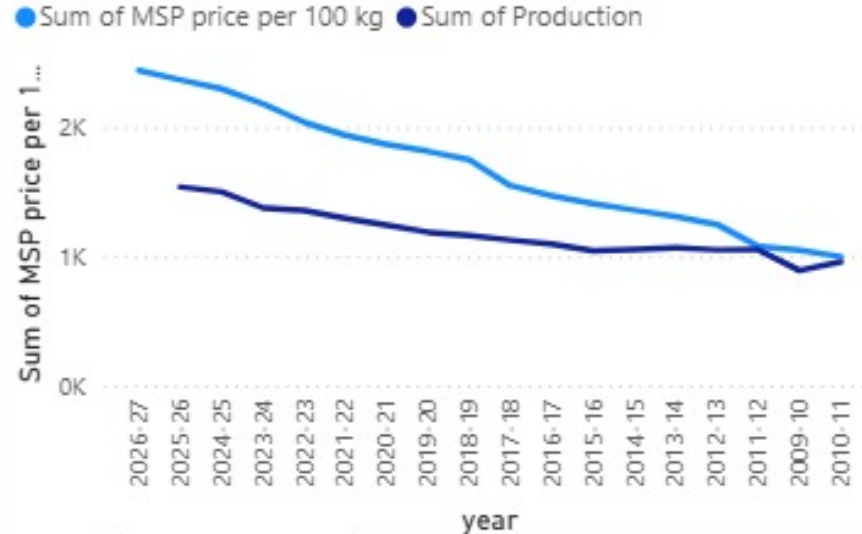
MSP by YEAR



AREA by YIELD



YIELD and MSP by year



Key Findings:

Production: Increased from 419 lakh tons to 1540 lakh tones, representing a CAGR of 2.74% over the past 50 years.

MSP: Increased from ₹340/Ton to ₹2441/ton, growing at 71.78% CAGR.

Production driver: [Area/Yield] has been the stronger contributor to production growth, with CAGR of 1.73% versus 0.58%.

Metric	1975-76	1985-86	CAGR
MSP	74	142	6.73%
Area	395	411	0.40%
Yield	1235	1552	2.31%
Productio	487	638	2.74%
Metric	1985-86	1995-96	CAGR
MSP	142	360	9.75%
Area	411	428	0.41%
Yield	1552	1797	1.48%
Productio	638	770	1.90%
Metric	1995-96	2005-06	CAGR
MSP	360	570	4.70%
Area	428	437	0.21%
Yield	1797	2102	1.58%
Productio	770	918	1.77%
Metric	2005-06	2015-16	CAGR
MSP	570	1410	9.48%
Area	437	435	-0.05%
Yield	2102	2400	1.33%
Productio	918	1044	1.29%
Metric	2015-16	2025-26	CAGR
MSP	1410	2369	5.33%
Area	435	528	1.96%
Yield	2400	2915	1.96%
Productio	1044	1540	3.96%

CAGR for every 10 year

Where are the major supply risks?

Largest declines (-23.04%) Year: 2002-2003

Largest increases (27.60) Year: 1983-84

Whether shocks came from area or yield : Yield is dominated

Is MSP growth keeping pace with productivity?

Yes, we get a continuous growth in **MSP** as the **Yield** increases

Historical production shows **moderate** volatility, with the largest supply shock occurring in **2002-03**, when production declined by **23.04%**, associated with an **8.24%** decline in area and a **16.11%** decline in yield. Meanwhile, MSP growth of **7.18%** has outpaced yield growth of **1.73%** over the long term.

Worst Production	Worst Decline Year	Worst MSP YoY	Worst Area YoY	Worst Yield YoY
-70.23043944	2002-03	0	-0.082405345	-0.161135161
-0.213754647	1979-80	0.117647059	-0.027160494	-0.19126506
-0.13963039	1976-77	0	-0.025316456	-0.118218623
-0.116322702	1982-83	0.060869565	-0.058968059	-0.058868502
-0.101814516	2009-10	0.166666667	-0.079120879	-0.024334252

Final Conclusion

- Production is expected to continue its long-term growth, but year-to-year fluctuations may remain.
- Yield improvement will remain important for sustaining production growth.
- MSP is expected to continue increasing, supporting farmer price realization.
- Key focus should be on improving productivity and reducing production volatility.

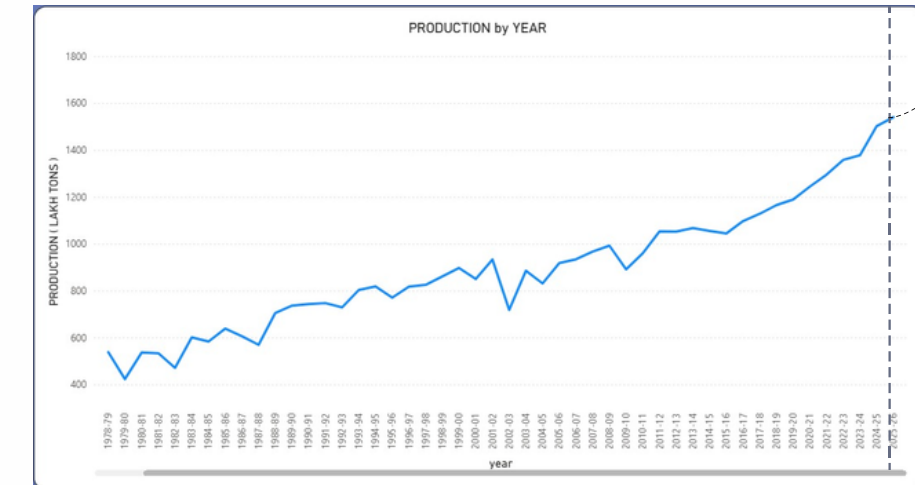
The outlook is positive, but sustained growth will depend more on **productivity** improvements than on expanding cultivated area.

Future Outlook

2026-27 PRODUCTION FORECAST

~1702
Lakh tonnes

Based on 2023-24 to 2025-26
CAGR



Thank You